

WULUNGURRU, Australia

WMO: 943210

Lat: 23.2657S

Lon: 129.3840E

Elev: 455

StdP: 95.97

Time Zone: 9.50 (AUC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	8.9	10.0	-14.5	1.1	22.1	-12.5	1.3	24.0	10.3	13.2	9.7	13.9	5.4	150	0.567

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.5	42.5	19.1	41.3	19.0	40.2	18.9	24.5	29.8	23.9	29.7	23.3	30.0	4.8	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.4	19.3	26.5	22.5	18.2	26.6	21.4	17.0	27.0	77.3	30.0	74.7	29.5	72.0	30.2	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.7	9.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.0	32.9	31.9	30.8	26.8	21.3	17.4	17.5	19.5	24.1	28.3	29.8	31.9
	DBStd	6.40	3.32	2.91	3.22	3.16	3.17	2.61	2.78	3.01	3.46	3.32	3.35	3.38
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	134	0	0	0	0	10	47	49	24	4	1	0	0
	CDD10.0	5829	709	613	643	503	349	223	233	295	423	567	593	678
	CDD18.3	2923	451	379	385	253	101	20	24	61	177	309	343	419
	CDH23.3	40103	7123	5778	5523	2787	726	113	214	554	1881	4063	4904	6436
	CDH26.7	24184	4814	3734	3372	1368	213	14	28	140	919	2357	3003	4222
Wind	WSAvg	5.3	5.3	5.2	5.5	5.4	5.2	5.1	5.0	5.1	5.2	5.3	5.5	5.3
Precipitation	PrecAvg	285	42	52	29	15	17	13	10	5	6	15	28	52
	PrecMax	746	127	350	206	102	92	73	63	41	70	77	81	206
	PrecMin	45	6	1	0	0	0	0	0	0	0	1	2	6
	PrecStd	153	28	65	45	24	24	15	16	8	12	18	21	40
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.3	42.5	41.5	38.0	32.4	28.5	28.7	31.8	36.6	39.8	41.7	44.3
		MCWB	19.8	18.8	18.7	18.7	15.4	14.7	13.6	14.5	15.7	16.9	18.1	19.4
	2%	DB	42.7	41.1	40.0	36.2	30.5	26.1	27.4	29.9	35.2	38.7	40.1	42.5
		MCWB	19.4	19.0	18.9	17.1	15.5	14.3	13.4	13.9	15.3	16.5	17.9	18.9
	5%	DB	41.3	39.9	38.7	34.9	29.0	24.4	26.0	28.2	33.7	37.7	38.9	41.0
		MCWB	19.5	18.9	18.8	16.9	15.2	13.3	12.8	12.8	14.8	16.3	17.7	18.9
	10%	DB	39.9	38.5	37.3	33.2	27.5	22.7	24.0	26.7	31.9	36.3	37.3	39.2
		MCWB	19.6	18.9	18.5	16.4	14.4	12.2	12.0	12.6	14.3	16.1	17.4	18.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.6	24.7	24.7	21.6	20.7	18.9	17.8	18.1	19.2	21.1	21.7	24.7
		MCDB	29.5	29.3	29.3	27.9	22.1	22.7	23.8	22.4	23.9	28.1	31.5	29.0
	2%	WB	24.8	24.1	23.7	20.5	18.8	16.8	16.1	16.0	17.6	19.8	21.2	23.6
		MCDB	30.4	30.1	29.2	27.7	23.7	22.2	23.2	25.0	27.0	27.7	31.4	29.1
	5%	WB	24.1	23.6	22.6	19.4	17.3	15.1	14.7	14.6	16.2	19.1	20.7	23.0
		MCDB	29.9	30.4	30.2	29.3	25.5	21.0	21.0	25.7	29.1	27.5	31.2	29.9
	10%	WB	23.4	22.9	21.8	18.2	16.1	13.8	13.1	13.2	15.1	18.2	20.1	22.3
		MCDB	29.8	30.6	30.6	28.8	24.7	21.0	21.5	25.2	29.9	27.8	30.7	30.9
Mean Daily Temperature Range	5% DB	MDBR	11.5	11.4	10.6	10.7	10.6	10.1	11.5	12.5	13.3	13.1	12.7	12.0
		MCDBR	12.7	12.4	12.1	11.9	11.8	11.6	13.6	14.6	14.9	14.9	13.9	13.3
		MCWBR	3.8	3.6	3.8	4.1	4.6	5.0	5.8	6.3	5.5	5.4	4.4	4.2
	5% WB	MCDBR	10.3	10.5	9.5	10.8	9.6	9.7	11.1	12.5	12.8	11.7	12.7	11.7
	MCWBR	2.8	3.0	3.0	3.6	4.4	4.6	5.1	6.1	5.3	4.0	3.4	3.2	
Clear-Sky Solar Irradiance	taub	0.360	0.359	0.347	0.328	0.310	0.294	0.300	0.310	0.350	0.357	0.366	0.365	
	taud	2.523	2.526	2.545	2.538	2.533	2.560	2.518	2.453	2.371	2.389	2.428	2.489	
	Ebn at Noon	983	970	956	932	910	907	912	936	937	961	971	980	
	Edh at Noon	113	110	104	96	88	82	88	103	121	125	123	117	
All-Sky Solar Radiation	RadAvg	7.62	7.19	6.40	5.67	4.66	4.21	4.50	5.56	6.56	7.30	7.59	7.34	
	RadStd	0.46	0.54	0.45	0.33	0.39	0.34	0.33	0.23	0.33	0.41	0.37	0.54	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page